

# Analyze Graphs of Relationships Between Two Variables

Lesson 9-2

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

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## My Notes Level 1 Support



### TODAY'S OBJECTIVES

**Content Objective:** I can use a table and a graph to determine and analyze the relationship between two variable quantities.

**Language Objective:** I can describe what a point on a graph represents using the words ordered pair, axis, and coordinates.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

graph

table of values

ordered pair

axis

coordinates



Tap any word to see what it means and a picture.

1

— A picture on a coordinate plane that shows how two quantities are related.

2

— A list of matched pairs of values that shows a relationship using numbers.

3

— Two numbers written as  $(x, y)$  that name one point on a graph. The independent variable comes first.

4

— One of the two number lines that frame a graph. Each axis is labeled with a quantity from the problem.

5

— The numbers in an ordered pair that tell how far to move along each axis to reach a point.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**What observations can you make about the data using the graph?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **graph** — A picture on a coordinate plane that shows how two quantities are related.

*gráfica — Un dibujo en un plano de coordenadas que muestra cómo se relacionan dos cantidades.*

- ☐ **table of values** — A list of matched pairs of values that shows a relationship using numbers.

*tabla de valores — Una lista de pares de valores que muestra una relación usando números.*

- ☐ **ordered pair** — Two numbers written as  $(x, y)$  that name one point on a graph. The independent variable comes first.

*par ordenado — Dos números escritos como  $(x, y)$  que nombran un punto en la gráfica. La variable independiente va primero.*

- ☐ **axis** — One of the two number lines that frame a graph. Each axis is labeled with a quantity from the problem.

*eje — Una de las dos rectas numéricas que enmarcan una gráfica. Cada eje se rotula con una cantidad del problema.*

- ☐ **coordinates** — The numbers in an ordered pair that tell how far to move along each axis to reach a point.

*coordenadas — Los números de un par ordenado que indican cuánto avanzar por cada eje para llegar a un punto.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

\_\_\_\_\_ is the independent variable, while \_\_\_\_\_ is the dependent variable. As the years go by, the interest earned \_\_\_\_\_, because each year adds \$\_\_\_\_\_ to the total.

*Di tu respuesta primero: Mi respuesta es \_\_\_\_\_ porque \_\_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The number of tickets is the independent variable, and the independent variable always comes first in an ordered pair, so the point is (3, 135), not (135, 3).

**Evidence:** A strong answer names tickets as independent and explains that independent always goes first, then checks the arithmetic ( $3 \times 45 = 135$ ). A weak answer just says 'because that's the order' without connecting it to independent/dependent.

**Reasoning:** This evidence connects the definition of graph to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- \_\_\_\_ is the independent variable, while \_\_\_\_ is the dependent variable. As the years go by, the interest earned \_\_\_\_, because each year adds \$ \_\_\_\_ to the total.

- My answer is \_\_\_\_.

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.

*Mi afirmación es \_\_\_\_.*

- I know because \_\_\_\_.

*Lo sé porque \_\_\_\_.*

- So \_\_\_\_.

*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
  - **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
  - **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*
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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

